UNIT – II

Fluid Kinematics, Dynamics & Pipe Flow

Subject : Fluid Mechanics & Hydraulic Machines (23A03402T)

> Syllabus: Types of flow, continuity equation, circulation & vorticity, streamline/pathline/streakline, stream & potential functions, flow net, source-sink-doublet-vortex, Euler's & Bernoulli's equations, momentum equation, pipe bend, Reynolds experiment, Darcy-Weisbach, minor losses, series/parallel pipes, TEL & HGL.

1. Fluid Kinematics – Introduction

Kinematics = study of fluid motion (velocity, acceleration, flow pattern) *without* considering the forces that cause it.

Types of flow

Classification	Type 1	Type 2
Time	Steady : $\partial()/\partial t = 0$ at a point	Unsteady : properties change with t
Space	Uniform : velocity same at all points ($\partial V/\partial s = 0$)	Non-uniform : V varies along flow
Nature	Laminar : orderly layers, Re small	Turbulent : eddies, mixing, Re large
Density	Incompressible : ρ = const (liquids)	Compressible : ρ varies (gases at high Mach no.)
Rotation	Irrotational : fluid particles do not spin	Rotational : particles rotate about own axes
Dimension	1-D : $V = f(s)$ (pipe average)	2-D : $u(x,y)$, $v(x,y)$; 3-D general

- Ex : flow through a pipe at constant Q = steady uniform ; flow through a tapering pipe at constant Q = steady non-uniform ; pipe with varying Q & constant diameter = unsteady uniform.

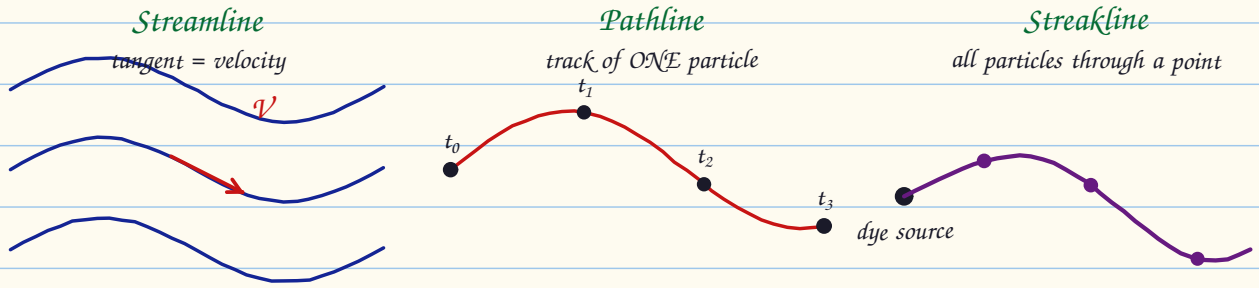
Acceleration of a fluid particle

Velocity $V = f(x,y,z,t)$. Acceleration in x-direction :

$$a_x = \partial u/\partial t + u\,\partial u/\partial x + v\,\partial u/\partial y + w\,\partial u/\partial z$$

- $\partial u/\partial t$ = local (temporal) acceleration – zero for steady flow. Remaining terms = convective acceleration – zero for uniform flow.

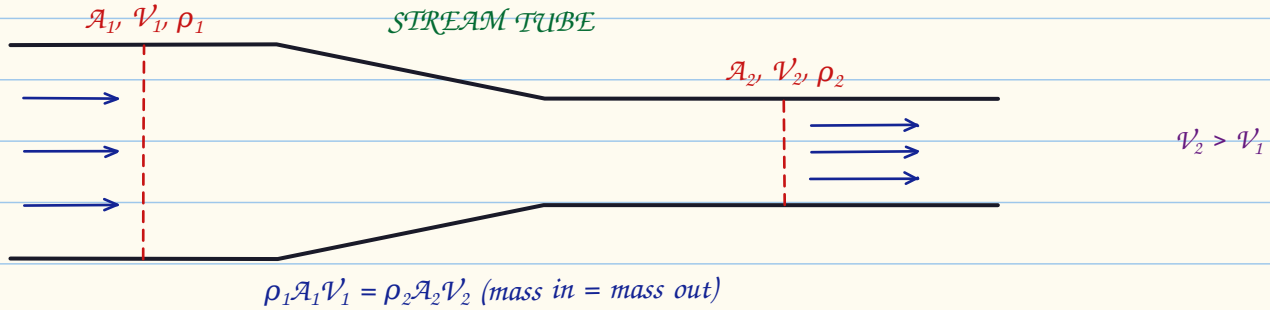
Streamline, Pathline, Streakline, Stream tube



- **Streamline** : imaginary curve whose tangent at every point gives the direction of velocity there. In steady flow streamlines never cross and no flow crosses them. Equation : $dx/u = dy/v = dz/w$.
- **Pathline** : actual track followed by one fluid particle over a time interval. **Streakline** : locus of all particles that have passed through a fixed point (dye/smoke filament).
- For steady flow all three coincide. **Stream tube** : tube formed by a bundle of streamlines through a small closed curve – behaves like a solid pipe wall.

2. Continuity Equation

Conservation of mass : mass entering a stream tube per second = mass leaving (steady flow).



$\rho_1 A_1 V_1 = \rho_2 A_2 V_2$ (mass in = mass out)

$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2 \text{ incompressible : } Q = A_1 V_1 = A_2 V_2$$

- For a branching pipe : $Q_1 = Q_2 + Q_3$.
- 3-D differential form (compressible, unsteady) : $\partial\rho/\partial t + \partial(\rho u)/\partial x + \partial(\rho v)/\partial y + \partial(\rho w)/\partial z = 0$

$$\text{Incompressible : } \partial u/\partial x + \partial v/\partial y + \partial w/\partial z = 0 \text{ (2-D : } \partial u/\partial x + \partial v/\partial y = 0)$$

Ex 1. A 300 mm pipe reduces to 150 mm. $V_1 = 2$ m/s. $Q = (\pi/4)(0.3)^2(2) = 0.1414$ m^3/s ; $V_2 = Q/A_2 = 0.1414/0.017671 = 8$ m/s (velocity $\propto 1/D^2$).

Ex 2. Is $u = 2x$, $v = -2y$ a possible incompressible flow? $\partial u/\partial x + \partial v/\partial y = 2 - 2 = 0$ ⇒ Yes, continuity is satisfied.

3. Rotation, Vorticity & Circulation

For 2-D flow, angular velocity of a fluid element about z-axis :

$$\omega_z = \tfrac{1}{2}\,(\partial v/\partial x - \partial u/\partial y)$$

- Vorticity $\zeta = 2\omega_z = \partial v/\partial x - \partial u/\partial y$ (twice the rotation).
- Circulation $\Gamma = \oint V \cdot ds$ (line integral of the tangential velocity round a closed curve). By Stokes' theorem $\Gamma = \zeta \times A$ (A = enclosed area).
- **Condition for irrotational flow** : $\omega_z = 0 \Rightarrow \partial v/\partial x = \partial u/\partial y$ (3-D : curl $V = 0$). Then $\Gamma = 0$ around any closed curve within the fluid.

Stream function ψ (2-D, incompressible)

Defined so that continuity is automatically satisfied :

$$u = \partial\psi/\partial y \quad v = -\partial\psi/\partial x$$

- ψ = constant along a streamline ; flow between two streamlines $Q = \psi_2 - \psi_1$ (per unit depth).
- Exists for any 2-D incompressible flow (rotational or not). For irrotational flow $\nabla^2\psi = 0$ (Laplace).

Velocity potential function φ

Exists only for irrotational flow :

$$u = -\partial\varphi/\partial x \; v = -\partial\varphi/\partial y$$

- φ = constant gives an equipotential line. Continuity $\Rightarrow \nabla^2\varphi = \partial^2\varphi/\partial x^2 + \partial^2\varphi/\partial y^2 = 0$ (Laplace equation).

Differences & relation between ψ and φ

Stream function ψ	Potential function φ
For 2-D flow only	Any dimension (3-D also)
Exists for rotational and irrotational flow	Exists only for irrotational flow
Satisfies continuity automatically	Satisfies continuity $\Rightarrow$ Laplace equation
ψ = const $\rightarrow$ streamline	φ = const $\rightarrow$ equipotential line
$u = \partial\psi/\partial y$; $v = -\partial\psi/\partial x$	$u = -\partial\varphi/\partial x$; $v = -\partial\varphi/\partial y$
Only for incompressible flow	Can be used for compressible flow

- Relation (Cauchy-Riemann) : $\partial\varphi/\partial x = -\partial\psi/\partial y$, $\partial\varphi/\partial y = \partial\psi/\partial x$. Slope of ψ-line × slope of φ-line = $-1 \Rightarrow$ *streamlines and equipotential lines are orthogonal.*

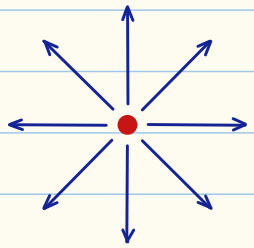
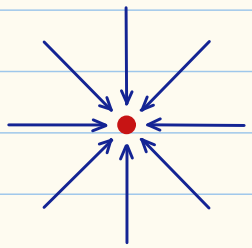
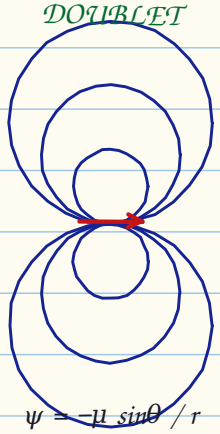
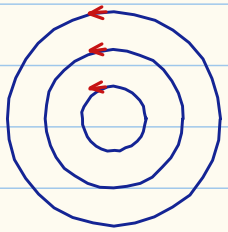
Ex 3. For $u = 2x$, $v = -2y$: $\psi = 2xy$ ($u = \partial\psi/\partial y = 2x$ □, $v = -\partial\psi/\partial x = -2y$ □) ; $\partial v/\partial x = 0 = \partial u/\partial y \Rightarrow$ irrotational, so φ exists : $\varphi = -x^2 + y^2$. (flow near a stagnation point).

Flow net

A grid of streamlines and equipotential lines intersecting at right angles forming (curvilinear) squares.

- Drawn graphically, by electrical/hele-shaw analogy or numerically. Closer lines $\Rightarrow$ higher velocity (and lower pressure).
- Uses : seepage under dams & weirs (flow through porous media), flow around bodies, through bends/contractions. Discharge $q = k H (N_f/N_d)$ for seepage.
- Limitation : only for ideal irrotational flow.

4. Basic Flows – Source, Sink, Doublet, Vortex

SOURCE	SINK	DOUBLET	FREE VORTEX
			
$\psi = (q/2\pi)\,\theta$	$\psi = -(q/2\pi)\,\theta$	$\psi = -\mu \sin\theta / r$	$v_\theta = \Gamma / 2\pi r$
q = outflow / unit depth	$v_r = -q/2\pi r$	source + sink, $d \to 0$	$\psi = -(\Gamma/2\pi) \ln r$

- ***Source*** *: flow radially outward from a point, strength q = discharge per unit depth. $v_r = q/(2\pi r)$, $v_\theta = 0$; $\varphi = -(q/2\pi) \ln r$; $\psi = (q/2\pi)\,\theta$. Sink is the negative of source.*
- ***Doublet*** *: a source and sink of equal strength brought infinitely close (distance $d \to 0$ with $q\,d = \mu$ held finite). $\psi = -\mu \sin\theta / r$. Streamlines = circles tangent to the axis.*
- ***Free (irrotational) vortex*** *: fluid circulates with $v_\theta = \Gamma/(2\pi r) \Rightarrow v\,r = const$; speed ↑ toward the centre ; pressure ↓ toward centre ; $\varphi = -\Gamma\theta/2\pi$.*
- ***Forced vortex*** *: fluid rotates like a rigid body, $v = \omega\,r$ (rotational, $\zeta = 2\omega$). Surface of liquid in a rotating cylinder : $z = \omega^2 r^2/2g$ (paraboloid).*
- *Superposition (add ψ values): uniform stream + source = Rankine half body ; source + sink = Rankine oval ; uniform stream + doublet = flow past a cylinder ; + vortex = cylinder with lift (Magnus effect).*

5. Fluid Dynamics – Forces & Euler's Equation

Forces acting on a fluid element :

- ***Surface forces*** *– act on the boundary : pressure force, shear (viscous) force, surface tension.*
- ***Body forces*** *– act on the whole mass : gravity, centrifugal, magnetic, electric.*
- *Others : inertia force (= mass × acceleration), compressibility force, turbulent force.*

Euler's equation of motion (along a streamline)

Consider a prism of length ds along a streamline : forces = pressure, weight component, no viscous force (ideal fluid). Newton's 2nd law : $p\,dA - (p + dp)\,dA - \rho g\,dA\,ds\cos\theta = \rho\,dA\,ds \cdot v\,dv/ds$.

$$dp/\rho + g\,dz + v\,dv = 0$$

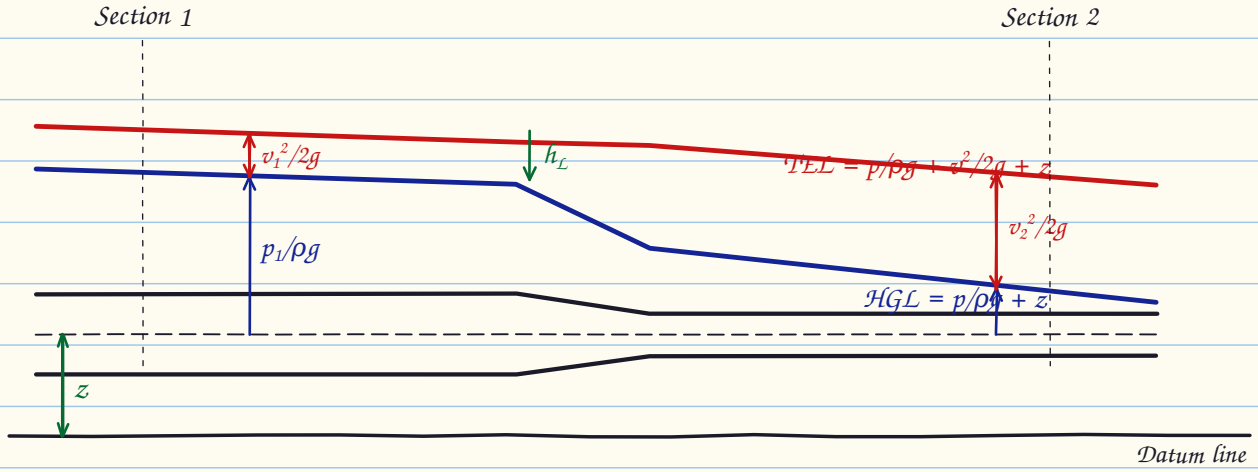
Assumptions : steady flow, along a streamline, ideal (inviscid) fluid, only gravity & pressure forces.

Bernoulli's equation

Integrate Euler's equation for an incompressible fluid (ρ = const) :

$$p/\rho g + v^2/2g + z = \text{constant}$$

- Terms (in metres of fluid) : pressure head $p/\rho g$; velocity (kinetic) head $v^2/2g$; datum (potential) head z.
- In energy per unit weight : pressure energy + kinetic energy + potential energy = const (total head H).
- Between two sections 1 and 2 with losses h_L : $p_1/\rho g + v_1^2/2g + z_1 = p_2/\rho g + v_2^2/2g + z_2 + h_L$. With pump head H_p added or turbine head H_t extracted add on the appropriate side.



- Limitations : ideal fluid, steady, incompressible, valid along a streamline (same constant for whole flow if irrotational), no heat/work transfer, correct for viscous flow only with h_L.

Applications of Bernoulli's equation

- **Venturimeter** (converging–throat–diverging) : $Q = C_d\, a_1 a_2 \sqrt{(2gh)} / \sqrt{(a_1^2 - a_2^2)}$; h = differential head = $x(S_{Hg}/S_{fluid} - 1)$ for a mercury manometer reading x.
- **Orifice meter** (cheaper, more loss, $C_d \approx 0.6$) ; **Pitot tube** : $v = C_v \sqrt{(2gh)}$ (stagnation pressure – static pressure). Also nozzles, siphons, orifices (Torricelli $v = \sqrt{(2gH)}$).

> Ex 4. Venturimeter 300 mm × 150 mm, oil-mercury differential $x = 0.2$ m, S_{oil}=1 (water) ⇒ $h = 0.2(13.6 - 1) = 2.52$ m. $a_1 = 0.0707$, $a_2 = 0.01767$ m² ; $Q = 0.98 \times (0.0707 \times 0.01767 / \sqrt{(0.0707^2 - 0.01767^2)}) \times \sqrt{(2 \times 9.81 \times 2.52)} = 0.126$ m³/s ≈ 126 l/s.

6. Momentum Equation & its Applications

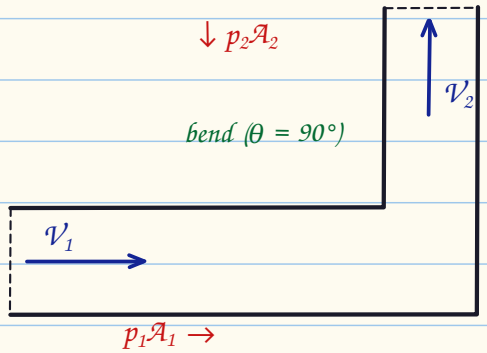
From Newton's second law, for a steady flow through a control volume :

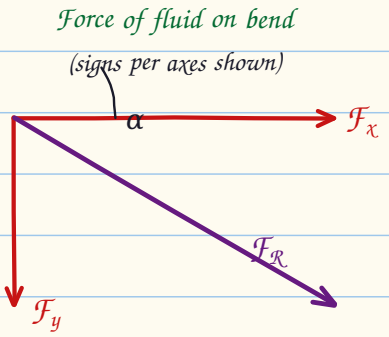
$$\Sigma F = \rho\, Q\, (V_{out} - V_{in}) \text{ (vector equation)}$$

F includes pressure forces on the end faces, body weight and the reaction of the solid boundary. Apply separately in x and y.

- Applications : force on pipe bends and reducers, force of a jet on plates and vanes, thrust on nozzles, hydraulic jump, propeller thrust, rocket motion.

Force on a pipe bend





Flow enters along x (V_1, A_1, p_1) and leaves at angle θ (V_2, A_2, p_2). Force exerted by the fluid on the bend :

$$F_x = \rho Q(V_1 - V_2 \cos\theta) + p_1 A_1 - p_2 A_2 \cos\theta$$

$$F_y = -(\rho Q V_2 + p_2 A_2)\sin\theta \quad F_R = \sqrt{(F_x^2 + F_y^2)}, \ \tan\alpha = F_y/F_x$$

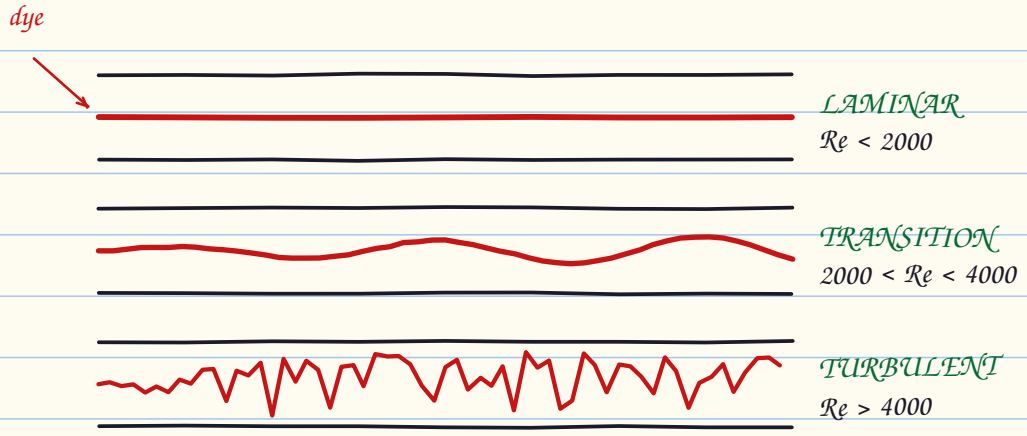
Procedure : (1) Q, V_1, V_2 from continuity ; (2) p_2 from Bernoulli ; (3) F_x, F_y ; (4) resultant and direction (the support/anchor block must resist it).

Ex 5. Horizontal 90° bend, 300 mm → 150 mm, $Q = 0.15\ m^3/s$, $p_1 = 200$ kPa. $V_1 = 2.12$, $V_2 = 8.49$ m/s ; Bernoulli ⇒ $p_2 = 200000 + 500(2.12^2 - 8.49^2) = 166.2$ kPa. $F_x = 1000(0.15)(2.12) + 200000(0.0707) = 14455$ N ; $F_y = -(1273 + 166232\times0.01767) = -4211$ N. $F_R = 15.06$ kN at $\alpha = 16.2°$ below the x-axis.

7. Closed Conduit Flow

Reynolds' experiment

Dye injected into water flowing in a glass tube at increasing velocity :



$$Re = \rho V D / \mu = V D / \nu \ \text{(ratio of inertia to viscous forces)}$$

- Pipe flow : $Re < 2000$ laminar ; 2000–4000 transition ; > 4000 turbulent. Critical velocity depends on temperature, roughness, disturbances.
- Laminar velocity profile parabolic, $V_{max} = 2\,V_{avg}$; turbulent profile flatter, $V_{avg} \approx 0.8\,V_{max}$.

Head loss – Darcy-Weisbach equation

Frictional (major) loss in a pipe of length L and diameter D :

$$h_f = f\,L\,V^2 / (2\,g\,D) = 8\,f\,L\,Q^2 / (\pi^2\,g\,D^5)$$

- f = Darcy friction factor (dimensionless). Laminar : $f = 64/Re$ (Hagen-Poiseuille : $h_f = 32\,\mu\,V\,L/(\rho\,g\,D^2)$).

- Turbulent smooth pipe : Blasius $f = 0.316\,Re^{-0.25}$ $(4000 < Re < 10^5)$. Rough pipes : $f = F(Re, \varepsilon/D)$ from Moody chart or Colebrook : $1/\sqrt{f} = -2\log(\varepsilon/3.7D + 2.51/(Re\sqrt{f}))$.
- Chezy's form : $V = C\sqrt{(m\,i)}$, $m = D/4$ (hydraulic mean depth), $i = h_f/L$. Also Fanning $f' = f/4$ $(h_f = 4 f' L V^2/2gD)$.

Ex 6. $L = 200$ m, $D = 0.1$ m, $Q = 0.02$ m³/s, $f = 0.02$: $V = 0.02/0.007854 = 2.546$ m/s ; $h_f = 0.02 \times 200 \times 2.546^2/(2\times9.81\times0.1) =$ 13.2 m.

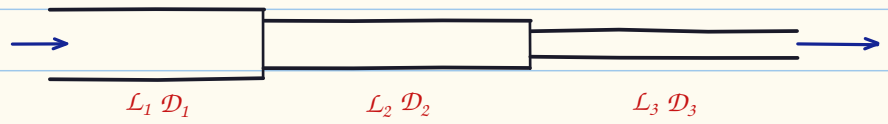
Minor losses

Losses due to change of section, direction or flow obstruction : $h_m = K V^2/2g$.

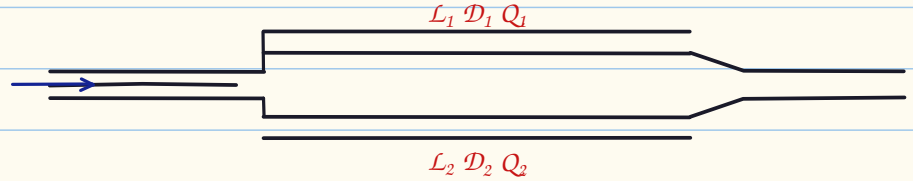
Fitting	Loss	Remark
Sudden enlargement	$h = (V_1 - V_2)^2/2g$	Borda–Carnot
Sudden contraction	$h = 0.5\,V_2^2/2g$ (approx.)	or $(1/C_c - 1)^2 V_2^2/2g$
Pipe entrance	$h = 0.5\,V^2/2g$	sharp-edged (bell-mouth 0.04)
Pipe exit to tank	$h = V^2/2g$	all KE lost
Bend / elbow 90°	$K \approx 0.4 – 0.9$	depends on R/D
Gate valve (open)	$K \approx 0.15 – 0.2$	globe valve ≈ 10
Gradual enlargement	$h = k(V_1 - V_2)^2/2g$	$k \approx 0.2 – 0.5$

- In long pipes $(L/D > 1000)$ minor losses are negligible; otherwise convert each to an equivalent length $L_e = K D / f$.

Pipes in series & parallel



SERIES : Q same
$H = h_{f1}+h_{f2}+h_{f3}$



PARALLEL : h_f same
$Q = Q_1 + Q_2 + \ldots$

- **Series** : same discharge, total loss = sum of losses : $H = \Sigma\, f L V^2/2gD$ + minor losses. Equivalent pipe (Dupuit) of length L and diameter D : $L/D^5 = L_1/D_1^5 + L_2/D_2^5 + L_3/D_3^5$.

- **Parallel** : same head loss in every branch, discharge adds : $Q = Q_1 + Q_2 + \dots$, and $h_{f1} = h_{f2} \Rightarrow Q_1/Q_2 = \sqrt{\left((D_1^5/L_1) / (D_2^5/L_2) \right)}$ (same f). Equivalent : $D^{5/2}/\sqrt{L} = \Sigma\, D_i^{5/2}/\sqrt{L_i}$.
- Pipe flowing from reservoir A to B through branching pipes (three-reservoir problem) – solved by trial for the junction piezometric head.
- Water hammer / siphon : pressure must stay above vapour pressure.

Total energy line (TEL) & Hydraulic gradient line (HGL)

- **HGL** = line joining $p/\rho g + z$ at each section (height of a piezometer column). **TEL** = $p/\rho g + v^2/2g + z$, always above HGL by $v^2/2g$.
- TEL slopes downward in direction of flow (friction); drops suddenly at fittings; rises only at a pump. For uniform pipe TEL ∥ HGL.
- If the pipe rises above the HGL, pressure there is negative (below atmosphere) – siphon action; risk of cavitation / air release.
- HGL for a pipe discharging freely to atmosphere ends at the pipe outlet centre-line.

8. Additional Solved Problems

P1 (Bernoulli). Horizontal pipe 200 mm → 100 mm, $p_1 = 150$ kPa, $V_1 = 2$ m/s (water, ideal). $V_2 = 2 \times 4 = 8$ m/s ; $p_2 = p_1 + \tfrac{1}{2}\rho(V_1^2 - V_2^2) = 150000 + 500(4 - 64) =$ **120 kPa**.

P2 (sudden enlargement). 100 mm → 200 mm, $Q = 0.03$ m³/s. $V_1 = 0.03/0.007854 = 3.82$ m/s ; $V_2 = 0.03/0.031416 = 0.955$ m/s ; $h_L = (3.82 - 0.955)^2/(2\times9.81) =$ **0.418 m**.

P3 (parallel pipes). Two pipes of equal length and same f : $D_1 = 0.2$ m, $D_2 = 0.1$ m, total $Q = 0.1$ m³/s. $Q_1/Q_2 = (D_1/D_2)^{5/2} = 2^{2.5} = 5.657 \Rightarrow Q_2 = 0.1/6.657 =$ **0.0150 m³/s** , $Q_1 =$ **0.0850 m³/s**.

P4 (orifice). Head 4 m above a 50 mm orifice, $C_d = 0.62$: $v = \sqrt{(2\times9.81\times4)} = 8.86$ m/s ; $Q = 0.62 \times 0.0019635 \times 8.86 =$ **0.0108 m³/s**.

P5 (Pitot). A Pitot-static tube in a water pipe shows a head difference of 0.15 m : $v = \sqrt{(2\,g\,h)} = \sqrt{(2\times9.81\times0.15)} =$ **1.72 m/s**.

P6 (laminar). Oil $\mu = 0.08$ Pa·s, $\rho = 900$ kg/m^3, $D = 50$ mm, $V = 0.5$ m/s : $Re = 900\times0.5\times0.05/0.08 = 281$ (laminar) ; $f = 64/281 = 0.228$; h_f per 100 m = $0.228\times100\times0.5^2/(2\times9.81\times0.05) = 5.8$ m.

Problem-solving checklist

- Draw the figure, mark sections 1 & 2, choose a datum. Write continuity first (V_2 from V_1), then Bernoulli with losses, then momentum if a force is asked.
- Pressures in Bernoulli may be all gauge or all absolute – never mix. Check Re before choosing f. Convert units (mm → m, l/s → m^3/s).
- Minor losses : find velocity in the pipe where the loss occurs (the smaller pipe for a contraction, $K V^2/2g$ with the downstream V).
- Sketch TEL and HGL after solving – TEL must fall along the flow except at a pump.

9. Quick Formula Sheet – Unit II

Topic	Formula
Continuity	$A_1V_1 = A_2V_2$; $\partial u/\partial x + \partial v/\partial y = 0$
Rotation / vorticity	$\omega = \tfrac{1}{2}(\partial v/\partial x - \partial u/\partial y)$; $\zeta = 2\omega$
ψ , φ	$u = \partial\psi/\partial y = -\partial\varphi/\partial x$; $v = -\partial\psi/\partial x = -\partial\varphi/\partial y$
Bernoulli	$p/\rho g + v^2/2g + z = const$
Momentum	$F = \rho Q(V_2 - V_1)$
Reynolds number	$Re = \rho VD/\mu$
Darcy-Weisbach	$h_f = fLV^2/2gD$; $f = 64/Re$
Minor loss	$h_m = K V^2/2g$
Series / Parallel	Σh_f ; $Q = \Sigma Q_i$

10. Important Questions

- Distinguish streamline, pathline and streakline. Derive the continuity equation in Cartesian co-ordinates.
- Define stream function and velocity potential function; give their differences and relation. Condition for irrotational flow.
- Explain source, sink, doublet and free vortex with equations. What is a flow net? State its uses.

- Derive Euler's equation and obtain Bernoulli's equation; state assumptions. Explain venturimeter with derivation.
- Derive the force exerted on a pipe bend using the momentum equation. Numerical problems.
- Explain Reynolds' experiment. Derive the Darcy-Weisbach equation. Write the Moody-chart based procedure.
- Minor losses in pipes; pipes in series and parallel; equivalent pipe; draw TEL and HGL for a pipeline.